

Simay Uygur

Curriculum Vitae (CV)

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SUMMARY

Fourth-year Computer Engineering student with strong foundations in algorithms, systems programming, digital design, and full-stack development. Experienced in backend engineering, AI-assisted systems, and hardware-level design through coursework and industry internships. Interested in backend development, distributed systems, and applied AI engineering.

HIGHLIGHTS

- Ranked top 25 in the Bilkent University CE cohort (CGPA 3.73).
- Industry experience at ASELSAN (REHIS) and multiple software internships.
- Strong background in algorithms, OOP, digital design, and low-level systems.

EDUCATION

Bilkent University - B.S. in Computer Engineering

2022 – Present

Ankara, Turkey

- Fourth-year student (1st semester).
- CGPA: **3.73**, ranked #25 among 190 students in the 2023 cohort.
- Advanced coursework in **Databases, Programming Languages, Software Engineering, Computer Architecture, and Operating Systems**.
- Designed and implemented **large-scale relational database systems** with normalization, indexing, triggers, and transaction management.
- Focused on backend system design, data consistency, concurrency, and performance trade-offs.
- Developed **full-stack web systems** using layered architectures, REST APIs, and role-based access control.
- Gained hands-on experience with **computer architecture** through MIPS Assembly programming and FPGA-based digital design.

Sanko Science and Technology High School

Graduated 06/2022

Gaziantep, Turkey

- Ranked #1706 nationally in the OSYM Math/Science exam.
- GPA 99.97, ranked #3 out of 90 students; awarded full scholarship.
- Participated in TUBITAK and MEF robotics competitions with STEM-focused project designs.

SKILLS

Programming Languages: Java, C++, Python, C#, TypeScript, JavaScript, SystemVerilog, MIPS

Frameworks & Tools: Spring Boot, FastAPI, React, .NET, Docker, Git, PostgreSQL, Firebase, SQLAlchemy

Core Knowledge Areas: Algorithms & Data Structures, Databases, OOP, REST APIs, Multithreading

Hardware & Systems: Arduino, FPGA (BASYS 3), Verilog/SystemVerilog, Computer Architecture

EXPERIENCE

ASELSAN - Software & Systems Test Intern

Radar and Electronic Warfare Systems (REHIS)

16/06/2025 – 11/07/2025

Ankara, Turkey

- **Technologies:** C#, .NET, TCP/IP Socket Programming, Client–Server Systems, Serialization
- Built a modular TCP communication simulator and supporting serialization infrastructure to streamline automated testing and debugging for internal defense-system workflows.

Akdoğan Technology - Software Development Intern

Ankara, Turkey

01/08/2025 – 29/08/2025

- **Technologies:** TypeScript, Python, FastAPI, React, PostgreSQL, Docker
- Contributed to a Dockerized FastAPI and React system by migrating frontend components to TypeScript, improving UI components, and backend development with PostgreSQL and SQLAlchemy.

Bilkent University - CS 101 Tutor

Ankara, Turkey

02/2024 – 05/2024

- **Technologies:** Java
- Supported first-year students in Java programming, debugging, and algorithmic problem-solving while reinforcing core software engineering concepts.

School Support Association (ODD) - Volunteer Science Teacher

Remote

01/2024 – 06/2024

- Delivered online science lessons to 6th-grade students using adaptive, inquiry-based teaching methods and received a Certificate of Appreciation for community contribution.

PROJECTS

Senior Design Project - Inquiry-Based Learning Platform (IIBLP)

2025 – Ongoing

- **Technologies:** Go, Tailwind CSS, Docker, LLM
- Co-developing an educational web platform that turns textbooks into interactive learning modules using LLM-generated questions, explanations, and backend workflow automation.

CS 353 - Sports League and Player Management System

Spring 2025

- **Technologies:** SQL, PostgreSQL/MySQL, Docker, Docker Compose
- Built a database system for managing sports leagues, tournaments, matches, and player data, with automated workflows, integrity constraints, and raw SQL-based operations. [GitHub Link](#)

CS 319 - TA Management System (Group Project)

Fall 2025

- **Technologies:** Spring Boot, TypeScript, HTML/CSS, MySQL, Apache Tomcat, Docker
- Developed a full-stack web application for managing TA workloads, proctoring assignments, leave requests, and administrative reporting with role-based access control. [GitHub Link](#)

CS 315 - Programming Languages Project (Custom Language with Lex & Yacc)

Spring 2025

- **Technologies:** C, Flex/Lex, Bison/Yacc, GNU Make
- Designed and implemented the front end of a custom programming language, including lexical analysis, syntax analysis, core language features, and automated testing. [GitHub Link](#)

CS 223 - UART Design and Implementation (FPGA Project)

2024

- **Technologies:** SystemVerilog, FPGA (BASYS 3), Digital Design, UART Protocol
- Designed and implemented a hardware-level UART communication system on FPGA with FIFO-based data buffering, parity handling, and display integration. [GitHub Link](#)